TIME-VAD: Text-Informed Magnitude Enhancement Feature Learning for Vehicle Accident Detection and Anticipation

Sumit Mishra, Medhavi Mishra, Pranjay Shyam, and Dongsoo Har, *Senior Member, IEEE*

***Abstract*—Vehicular accidents pose a substantial risk to drivers, underscoring the persistent and vital need for heightening safety measures. Early accident anticipation mechanisms are imperative for proactive measures, while detection accuracy is pivotal for prompt response and effective post-accident mitigation. Accurate and early anticipation of accidents for automated driving assistance systems in vehicles or CCTV in cities remains a complex task due to the intricate spatial-temporal interactions within traffic videos. This study presents text-informed magnitude enhancement in contrastive multiple-instance feature learning for vehicle accident detection and anticipation (TIME-VAD). Text is a better representative of concepts when compared to images in video, thus multi-modal learning is suitable. Also, the traditional assumption about feature magnitude of accidents and normal frames in magnitude based multiple-instance learning using weak supervision may not hold. This has led to the development of a novel weak-supervised learning strategy involving magnitude enhancement from textual concepts. For a better frame-level perception of accident risks in videos, dynamic temporal attentions are refined using the proposed dilated temporal conv-attention (DTCA) block. In-depth component-level analysis is performed to showcase the model's efficacy while elucidating its operational mechanisms. Evaluation is conducted on three benchmark datasets, considering both earliness and accuracy-related metrics. Extensive experiments demonstrate that our TIME-VAD model outperforms the existing models. Compared to the previous top-performing supervised model achieving 84.7% accuracy, TIME-VAD achieves a 94.44% accuracy (measured by ROC-AUC) on the largest DoTA dataset. Notably, our model also excels in measuring how early it detects accidents compared to previous methods. The code will be released on https://github.com/sumitmishra209/TIME-VAD**

***Index Terms*—Artificial surveillance, multi-modal learning, accident detection, real-time systems.**

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Sumit Mishra is with the Robotics Program, Korea Advanced Institute of Science and Technology, Daejeon 34141, South Korea (e-mail: sumitmishra209@kaist.ac.kr).

Medhavi Mishra and Dongsoo Har are with the CCS Graduate School of Mobility, Korea Advanced Institute of Science and Technology, Daejeon 34141, South Korea (e-mail: medhavi132@kaist.ac.kr; dshar@kaist.ac.kr).

Pranjay Shyam is with Amazon Fulfillment Technologies and Robotics, Seattle, WA 98109 USA (e-mail: pranjayy@amazon.com).

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I. Introduction

SURVEILLANCE systems, prevalent due to the widespread deployment of CCTV cameras, pose a significant challenge: the inherent limitations in human monitoring capabilities [1]. However, the advent of artificial intelligence (AI) offers a transformative solution by automating accident detection and anticipation. A fundamental aspect of effective accident detection and anticipation is understanding anomalies. It is imperative to comprehend that anomalies pave the way to accidents, rendering accidents as anomalies when juxtaposed with regular traffic patterns. For example, a person suddenly starting to run fast on a heavy traffic road with approaching vehicles can be an anomaly that could lead to an accident. Many anomaly detection datasets include vehicle accidents as one class among others, such as explosions, acts of crime, and acts of violence. However, vehicle accident detection tasks require different treatments than other anomaly classes. The challenges posed by accidents stem from their inherent unpredictability and context-dependent manifestations. Each accident class varies significantly, presenting a multifaceted hurdle that does not align neatly with conventional clustering, classification, or regression paradigms. This necessitates adaptations from these areas for accident detection. Recent surveys highlight the growing importance of vision-based traffic accident detection and anticipation systems [60], with approaches ranging from cognitive modeling [58] to ego-view understanding [59]. Some works address anomalies along with vehicle accidents, and some focus solely on vehicle accidents, highlighting the unique requirements of accident detection tasks [4].

Video accident detection and anticipation by leveraging deep learning (DL) and other classes of AI emerge as a promising avenue due to their ability to mine semantic patterns from automatically learned features. Through end-to-end optimization for feature extractions and accident assessments, diverse DL models exhibit the capacity to anticipate accidents across multiple levels spanning frame, temporal-spatial, and pixel domains and concurrently detect motion. The accident anticipation task involves identifying accidents before they happen, while detection focuses on recognizing the start of an accident event. As illustrated in Fig. 1, our proposed framework integrates both detection and anticipation capabilities through a unified architecture. The DL model's output probability when compared with a threshold determines if an

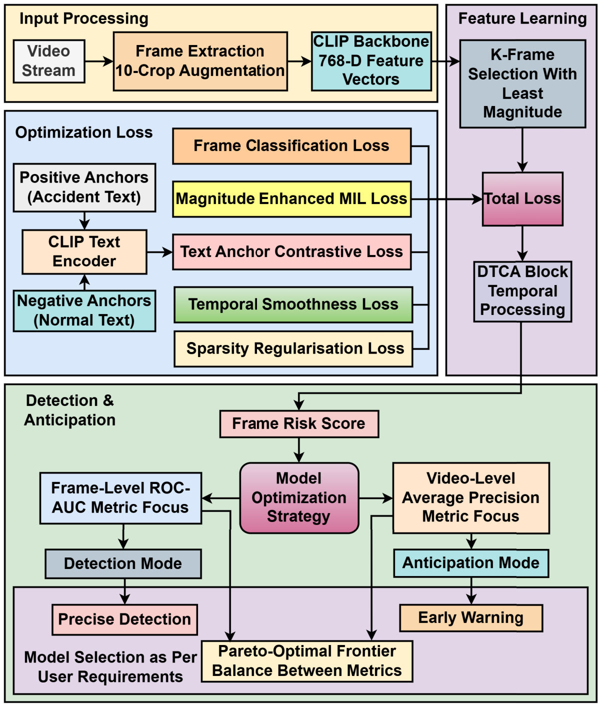

Fig. 1. TIME-VAD framework architecture for unified accident detection and anticipation.

event in a frame is classified as a near accident or an accident. For both tasks, feature extraction and further processing are crucial steps in any DL model workflow. Therefore, we address these tasks together as accident detection based on earliness and accuracy. Existing efforts predominantly focus on unsupervised accident detection [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [18] that primarily model normal behavior and identify deviations from it without utilizing labeled accident data. They often rely on factors like motion trajectories or low-level features extracted from image patches to detect and anticipate accidents, exhibiting strength in simplifying the accident detection problem by generalizing to diverse accident distributions. While effective in certain scenarios, existing work may struggle in crowded scenes or fail to generalize in complex environments. Other limitations involve high false-positive rates, poor adaptation to environmental changes, and the inability to leverage readily available accident examples that can give prior knowledge about accidents. These constraints may hinder DL models' practical utility [19].

Building a fully supervised classification approach is difficult due to the heterogeneous nature and unknown manifestations of accidents until occurrence and the challenge of full supervision for all accident classes [19]. To overcome these challenges, self or weak supervision is explored to improve accident detection accuracy. Improved detection accuracy can be achieved through semi-supervised methods that extract normal patterns or by employing limited or weak supervised techniques such as multiple instance learning (MIL), which utilizes easy-to-annotate video-level labels for training [20]. Recent advancements in text and image-based contrastive learning have showcased notable efficacy with the availability of DL models, like Contrastive Language-Image Pre-Training (CLIP) [22], pretrained with large databases. These models offer features that can be directly utilized or further refined through contrastive methods. However, the challenge in training unsupervised and supervised accident detection frameworks demands a distinct approach to feature refinement. Therefore, a better technique different from the traditionally applied supervised [23] or unsupervised contrastive learning [24] is required in accident detection. Textual representations often articulate concepts more explicitly than images, prompting the exploration of employing contrastive supervision using text. Leveraging text-based signals might offer a more comprehensive and generic portrayal of accidents. Given the same concept for image and textual data, it is anticipated that image features corresponding to text features should exhibit similar trends in magnitude. This correlation should be a pivotal consideration in structuring the framework for MIL. From this viewpoint, the present work seeks to tackle these intricate challenges by proposing a novel approach leveraging text anchor-based contrastive weak-supervised feature enhancement within a MIL framework (Fig. 2), which has been tailored specifically for vehicle accident detection (VAD). Following are the main contributions of this work.

- A novel framework, Text-Informed Magnitude Enhancement Feature Learning for Vehicle Accident Detection (TIME-VAD) is introduced. It leverages CLIP features for contrastive fine-tuning using text-based anchors, proving more effective in accident detection by harnessing the explicit concepts articulated in textual data.
- Upon analyzing accident and normal traffic concepts in textual form, it is observed that, contrary to traditional assumptions, the magnitudes of normal CLIP features are higher than those of accident features. Consequently, a reversal of magnitudes is implemented to enhance learning by using weak-supervised magnitude-based MIL.
- To refine temporal features, a dilated temporal conv-attention (DTCA) block is proposed, dynamically attending to salient temporal segments. This innovative block aims to enhance feature representation in the temporal domain.
- While accident detection and anticipation are related, previous studies have addressed them separately, focusing on either timeliness or correctness. Our study achieves superior performance for both tasks independently and provides pragmatic results when combined.
- Comprehensive analyses and experiments are conducted across three diverse datasets, evaluating metrics related to the correctness and timeliness of accident frame detection. The proposed framework can showcase superior performance as compared to state-of-the-art models, underscoring its efficacy in accurately and promptly detecting traffic accidents.

Our work primarily focuses on vehicle accident detection, employing AI for real-time identification. The swift responses enabled by AI not only expedite aid to accident scenes, promising a significant reduction in casualties but also prioritize the improvement of advanced driving assistance systems (ADAS) [2]. By swiftly detecting accidents and promptly notifying drivers via ADAS or using it in collision-avoidance assistance systems, such as automatic emergency braking and lane-keeping assistance, vehicular safety could be bolstered [3]. Consequently, our work resides at the juncture of pressing societal needs and technological innovation, seeking to harness AI's potential for accident detection in surveillance and ADAS applications. The rest of this article is organized as follows. Section II reviews the existing accident detection works as well as other related tasks. Our proposed method is introduced in Section III. The comparison and component analysis experiments on the benchmarks are given in Section IV. Finally, Section V concludes the paper.

II. Literature Review

A. Weakly Supervised Anomaly and Accident Detection

Weakly supervised video anomaly detection capitalizes on using both normal and abnormal data with varying levels of supervision, leveraging video-level annotations to enhance anomaly detection [25], [26], [45], [46]. Unlike general anomaly detection, traffic accident detection presents unique challenges requiring temporal reasoning about evolving dangerous situations and early detection capabilities [60], [63]. The DL-based MIL has emerged as a key methodology within weakly supervised techniques [20], [27], [28], [29], [30], [31]. However, challenges persist in accurately predicting accident frames due to the complexity of the context leading to accidents, which prompts the exploration of novel adaptations. Recent advancements using DL encompass temporal context aggregation [28] and the integration of attention mechanisms [30] in the classification module. These aim to better discern abnormal events amidst normal video sequences. Encoder-based methods have also gained prominence, training both feature encoders and classifiers to refine predictions and optimize feature representations more efficiently. These strategies incorporate pseudo-label generation and self-training schemes [27], [28]. Recent advancements include self-supervised consistency learning [61] and text-driven temporal modeling [65] for improved detection accuracy.

B. Contrastive Learning for Anomaly and Accident Detection

Among recent anomaly detection techniques, the integration of contrastive learning methods has drawn significant attention. Some approaches, e.g., [32], [33], [34], directly integrate contrastive learning, showcasing novel models leveraging contrastive attention modules with classifiers. These methods refine anomalous predictions through attention mechanisms, significantly improving frame-level receiver operating characteristic - area under the curve (ROC-AUC) and highlighting their effectiveness in weakly-supervised anomaly detection scenarios. However, while these methods offer potential solutions, challenges persist. For instance, the direct application of contrastive learning exhibits limitations, motivating efforts to pair it with MIL for enhanced anomaly detection [36]. Hence, contrastive self-supervised learning within the realm of MIL is explored to overcome challenges posed by class imbalances [35]. Instance-level pseudo labels derived from bag-level annotations that substantially enhance instance-level accuracy across medical image datasets have been introduced. Similar to supervised and unsupervised learning, supervised and unsupervised contrastive learning face the same constraints, prompting further innovative strategies. Recent approaches leverage various techniques including two-stream networks [64] and monocular depth-enhanced 3D modeling [62] to improve detection capabilities. Previous studies have investigated weakly supervised anomaly detection based on temporal cues and feature discrimination [34], [36]. For traffic accident analysis, text-informed approaches [16], [65] have shown superior performance over image-based anchors in representing complex traffic concepts such as "accident scene" and "normal traffic", which are better expressed textually.

C. Multimodal Traffic Accident Detection and Anticipation

Recent advances in traffic accident detection have evolved beyond general anomaly detection to address domain-specific challenges through multimodal approaches. While multimodal anomaly detection traditionally leveraged audio-visual data [37], traffic scenarios present unique constraints. Extracting external road audio from enclosed vehicle environments or obtaining clear audio in CCTV feeds poses significant challenges, making textual input a viable alternative for multimodal learning [16], [65]. Long video event retrieval and description methods [38] employ superframe segmentation to mitigate computational complexity, underscoring the need for language-based methods in processing vast accident-related video content.

Contemporary traffic accident analysis approaches encompass cognitive modeling [58] for causal understanding and ego-view analysis [59] for driver-centric perception. Recent surveys [60] emphasize the shift toward vision-based systems with multimodal integration. Text-based anomaly detection [17] has emerged using weakly supervised MIL while considering classification probability on video level, while systematic reviews [63] identify real-time processing and early detection as critical requirements. Advanced architectures include self-supervised consistency learning [61], two-stream networks [64], and depth-enhanced 3D modeling [62]. Unlike general approaches focusing on classification probability for video, our method addresses frame-level vehicle accidents through feature magnitude enhancement within a text-informed multimodal framework, leveraging semantic understanding for improved discrimination in vehicular contexts.

III. Methodology

In this section, the problem statement is presented and the proposed method TIME-VAD is described in detail. The pipeline to train and optimize the TIME-VAD is also presented.

A. Problem Statement

In the MIL-based method, which is a weakly supervised VAD, the training videos are labeled solely at the video level. Videos with accidents are referred to as an abnormal video and is labeled as 1 (positive) while videos without any accidents are referred to as normal and is labeled as 0 (negative). Thus only video label $Y \in \{0, 1\}$ is provided. Let T denote the number of snippets (frames) in a video, K the number of selected features for MIL learning, m the separation margin, D the feature dimension (768 for CLIP), and L, M the numbers of positive and negative text anchors respectively. The operations largest-K and least-K select the K features with highest and lowest magnitudes respectively. The MIL-based methods treat the video as a bag and each snippet as an instance, defining a positive video as a positive bag of features $\mathcal{B}_a = (a_1, a_2, \ldots, a_T)$ and a negative video as a negative bag of features $\mathcal{B}_n = (n_1, n_2, \ldots, n_T)$. In this context, any video $V = \{v_i\}_{i=1}^{T}$, comprising of T snippets. The objective of VAD is to learn a function f_θ that maps snippet features either from $\mathcal{B}_a$ or $\mathcal{B}_n$ to their accident scores or probability within the range of 0 to 1 such that the following assumption

$$\max_{i \in \mathcal{B}_a} f_\theta(a_i) > \max_{i \in \mathcal{B}_n} f_\theta(n_i) \tag{1}$$

is valid. Formulating VAD as a classification task for outputting accident score, a MIL ranking objective function and MIL ranking loss are proposed in [20]. The objective function as shown in Eq. (1) aims to ensure that the snippet with the highest classification score in the positive bag ranks higher than the one with the highest classification score in the negative bag. The loss based on this objective is given as

$$\mathcal{L}(\mathcal{B}_a, \mathcal{B}_n) = \max\left(0, \max_{i \in \mathcal{B}_n} f_\theta(n_i) - \max_{i \in \mathcal{B}_a} f_\theta(a_i)\right) \tag{2}$$

To maintain a significant margin between positive and negative instances, a hinge-based ranking is used. This loss, when considering the maximum probability of 1 as a hinge, is given as

$$\mathcal{L}(\mathcal{B}_a, \mathcal{B}_n) = \max\left(0, 1 - \max_{i \in \mathcal{B}_a} f_\theta(a_i) + \max_{i \in \mathcal{B}_n} f_\theta(n_i)\right) \tag{3}$$

MIL-based methods in accident detection encounter inherent challenges that impede their efficacy. These methods face ambiguities in achieving a high probability score for the actual accident snippet of a video. This results in potentially misaligning the highest accident score snippet with the actual accident snippet. Issues arise when at the start of optimization, f_θ predicts normal snippets in typical videos with the highest classification score. This potentially reinforces errors during training, hindering training convergence, and blurring the distinction between normal and accident behavior. Moreover, videos with multiple accident snippets may lack sufficient instances for robust learning. This is because only the highest classification score is used, impacting the training process.

To overcome MIL challenges, the snippet feature magnitudes (L1 or L2 norm), instead of classification scores, are used in training by robust temporal feature magnitude (RTFM) [39] to differentiate between normal and abnormal snippets. The RTFM assumes that the low magnitude of the snippet's feature signifies normal behavior, while higher magnitudes indicate accidents. In addition, the RTFM learning considers K snippets instead of just one. The learning enforces a criterion where the mean of the largest-K feature magnitudes in abnormal video snippets is higher than that of the mean of the largest-K feature magnitudes in normal video snippets. For better separability, a margin m is considered between the largest-K feature magnitudes of abnormal video snippets and the largest-K feature magnitudes of normal video snippets. This criterion provides a more robust metric for accident detection. Based on the largest-K feature magnitudes, the MIL loss function in RTFM is given by

$$\mathcal{L}(\mathcal{B}_a, \mathcal{B}_n) = \max(0, m - d_\theta) \tag{4}$$

where

$$d_\theta = \frac{1}{K}\sum \text{largest-K}(\|a_i\|_2) - \frac{1}{K}\sum \text{largest-K}(\|n_i\|_2)$$
$$\max(\|a_i\|_2) > \max(\|n_i\|_2).$$

Also, the RTFM approach enforces alignment between the classification score of the largest-K magnitude snippets and their respective videos. This involves training a binary cross-entropy-based classification loss function, $\mathcal{L}_f(f_\theta, y)$ where $y \in (1, 0)$ such that y for the largest-K magnitude features of snippets from an accident video is 1 and y for largest-K magnitude features of snippets from a normal video is 0. The y for largest-K magnitude features of snippets from a normal video is 0 as there is no accident snippets in normal videos. The variable f_θ is a function of x, which is the largest-K magnitude of feature embeddings from each of $\mathcal{B}_a$ and $\mathcal{B}_n$. By jointly training normal and abnormal samples in both feature and scoring spaces, the method aims to ensure a comprehensive learning process. This video label classification loss for the largest-K features of snippets from accident and normal videos is given as

$$\mathcal{L}_f(f_\theta, y) = \sum_{x \in \text{largest-K}} -\big(y \log(f_\theta(x)) + (1-y)\log(1 - f_\theta(x))\big) \tag{5}$$

B. Time-VAD Framework

In the existing RTFM method, several limitations are identified, prompting the development of the TIME-VAD framework shown in Fig. 2. The RTFM framework involves challenges in distinguishing positive snippet embedding from negative snippet embedding, as their classification scores kept fluctuating while training. To address this, the TIME-VAD framework incorporated contrastive text anchor-based training to improve the distinction between positive and negative snippet embeddings. Anomaly detection studies [18], [39], [45], [46] used preprocessed features such as optical flow [43], and object bounding boxes [42] along with features from RGB or Gray pixels like I3D [40] or C3D [41]. Incorporating additional features along with I3D and C3D image-based backbones improves results but preprocessing additional features is computationally heavy as well as time-consuming. TIME-VAD employs the CLIP feature extraction backbone, capitalizing on its ability to comprehend multimodal (text and image) information, unlike conventional image-based backbones. Thus, a richer representation of data is enabled by using only the CLIP RGB pixel features. This is particularly beneficial in scenarios requiring nuanced contextual understanding across modalities. Furthermore, the assumption that accident frames exhibit higher magnitudes than normal frames did not hold

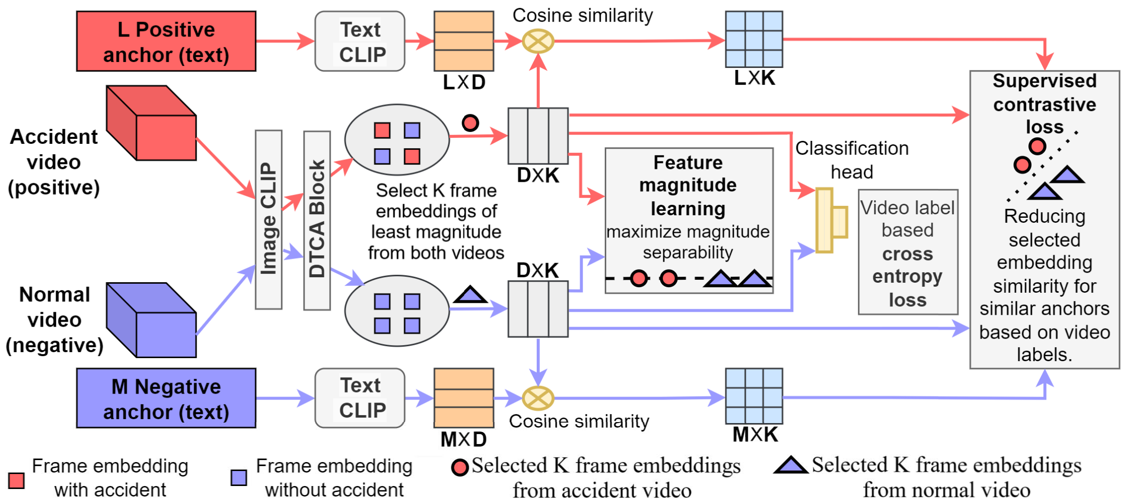

Fig. 2. TIME-VAD feature learning framework with color-coded processing paths: red for accident-related (accident video, positive anchors, accident frames) and purple for normal-related (normal video, negative anchors, normal frames) components. Accident videos contain both accident frames (red) and normal frames (purple).

universally. This is observed particularly in the case of CLIP, where this pattern is reversed. Hence, The TIME-VAD has enhanced the magnitude using MIL accordingly. Additionally, to refine features before classification, a DTCA block is introduced in TIME-VAD.

1) Clip Feature Extraction and Text Anchor Analysis: The TIME-VAD integrates the CLIP feature extraction backbone, deviating from conventional image-based models like I3D and C3D. CLIP's unique strength lies in its capacity to understand multimodal information, associating images with their respective texts. Unlike standard image-centric models, CLIP trained through contrastive learning comprehends the nuanced relationships between images and text. This broader comprehension enables CLIP to encapsulate not only visual data but also semantic and contextual information, significantly enhancing accident detection capabilities. The CLIP model [vit-large-patch14-336] [47], serving as TIME-VAD's feature extraction backbone, undergoes training on vast datasets, incorporating distinct image-description pairs. This model employs an image encoder built around the ViT architecture [48], which features variations in layer count, patch size (14×14), and input image dimensions (336×336 pixels). Simultaneously, its text encoder employs a transformer architecture [49] with an expanded layer count, amplifying its grasp of textual contexts for a more comprehensive understanding. Both texts and images have 768 dimension feature vectors.

Positive and negative text anchors can be either manually designed or learned. However, hand-crafted text prompts do not require additional training procedures and still achieve decent results [15]. Hence, we have manually designed multiple positive and negative text anchors. The text anchors are not exhaustive, as the fundamental purpose of weak supervision is acknowledging that exhaustively listing all possible accident scenarios is impractical and unnecessary. To develop effective anchors covering the accident types represented in textual descriptions of accidents and normal driving events from the W3AL database [73] GPT4, along with manual refining, is used to generate representative text anchors. During training, we considered the average effect for the corresponding set of anchors accordingly. A magnitude distribution analysis is conducted on 19 negative and 27 positive text anchors listed in Table I. The analysis, illustrated in Fig. 3, showcases anchor frequency against magnitude. For positive anchors, the mean

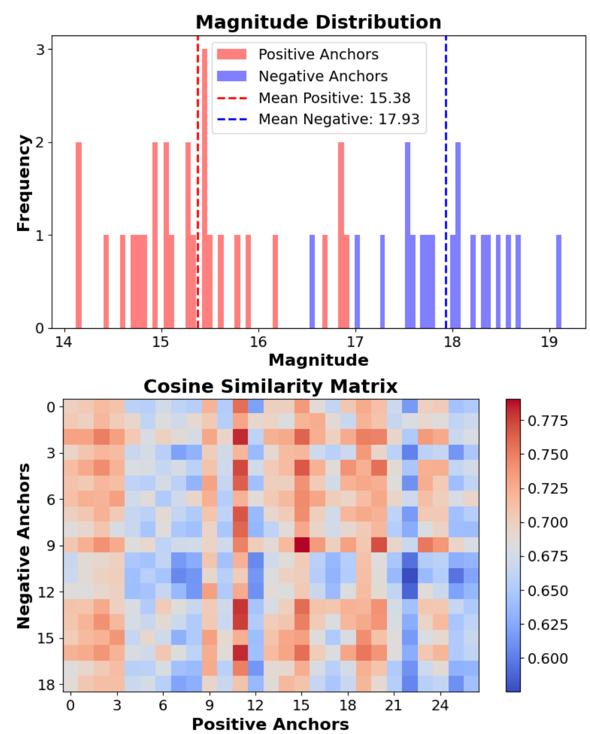

Fig. 3. CLIP text anchor analysis: Magnitude distribution histogram shows embedding frequency per magnitude bin, with accident anchors having lower mean magnitude than normal anchors. The cosine similarity matrix between accident and normal anchors, lying in the range of 0.575-0.8, reveals difficulty in distinguishing events in the textual modality.

TABLE I
TEXT FOR POSITIVE AND NEGATIVE ANCHORS USED IN TIME-VAD TRAINING

Negative anchors
"Smooth traffic flow", "Steady movement of vehicles", "Conventional traffic", "Routine commute", "Ordinary traffic conditions", "Regular traffic patterns", "Smooth driving", "Standard traffic flow", "Usual traffic congestion", "Typical road conditions", "Predictable rush hour", "Daily commute routine", "Consistent traffic patterns", "Common road congestion", "Ordinary daily traffic", "Regular travel conditions", "Familiar traffic flow", "Steady traffic pace", "Typical morning commute"
Positive anchors
"Scene with crash", "Scene with accident", "Scene with collision", "Scene with impact", "Scene with wreck", "Scene with damage", "Scene with pileup", "Scene with fender-bender", "Scene with rollover", "Scene with overshoot", "Scene with hazard", "Scene with traffic incident", "Scene with vehicle damage", "Scene with smash", "Scene with mishap", "Scene with road incident", "Scene with vehicle collision", "Scene with head-on collision", "Scene with intersection accident", "Scene with incident", "Scene with road accident", "Scene with collision impact", "Scene with wrecked vehicle", "Scene with roadway collision", "Scene with highway accident", "Scene with rollover accident", "Scene with side-impact collision"

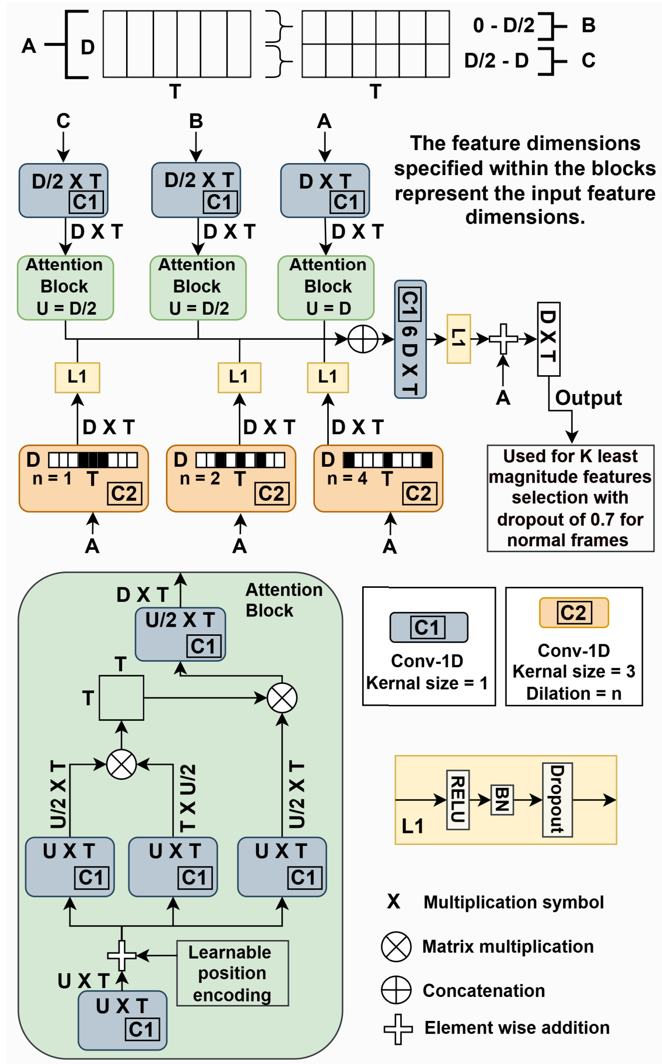

Fig. 4. Schematic of Dilated Temporal Conv-Attention (DTCA) block. The block consists of four main components with unique color coding: attention blocks (green) for global temporal context, C1 blocks (blue) for dimension adjustment, C2 blocks (orange) for dilated convolutions with varying dilation rates, and L1 blocks (yellow) for feature processing. The input feature dimensions are mentioned within the blocks and the corresponding output feature dimensions are mentioned along the output lines extending from the block and connecting to other blocks. For Conv-1D (C1), the channel dimensions are adjusted as per the required output feature dimension.

magnitude was registered at 15.38, contrasting with 17.93 for negative anchors. This disparity challenges the assumption that accident frames always exhibit higher magnitudes than normal frames. Additionally, examining cosine similarity, which is the dot product of unit vectors between positive and negative text anchor vectors (as shown in Fig. 3), revealed a range spanning from 0.575 to 0.8. This exploration highlighted the challenges in discerning between "accident scene" and "normal traffic". This was particularly evident in textual representations, which are albeit clearer compared to their visual counterparts. These complexities underscore the intricate nature of the problem domain, especially in visual conceptualizations of such scenarios.

2) TIME-VAD: TIME-VAD introduces an innovative architecture leveraging the CLIP backbone [22], harnessing the power of multimodal information. Initially, normal and accident videos are simultaneously input, and each frame from both videos undergoes 10-crop data augmentation for robust data. The CLIP backbone [vit-large-patch14-336] [47] extracts features from each frame of the video and encodes them in a 768-dimension (D) vector. Employing the DTCA block refines these features in the temporal domain to consider a given video rather than the frame images of that video independently. This enhances the features' representation for temporal context understanding.

Subsequently, from both the accident and normal videos, K frame feature embeddings with the least magnitude are selectively extracted. These frame feature embeddings that are processed through the proposed architecture undergo three primary loss functions. The first loss involves employing features with the least-K magnitude from both videos, emphasizing magnitude-enhanced MIL. The second loss employs features with the least-K magnitude, from both videos, in a classification head. Here, the classification probability of K features associated with the accident video is driven towards 1, while those of the normal video features are pushed towards 0, intensifying the discernment between normal and accident instances. Furthermore, the third loss integrates contrastive textual anchor-based training. Utilizing L positive and M negative text anchors features that have dimension D are computed using CLIP. For the least-K magnitude features extracted from the accident and normal videos, cosine similarity calculations are performed with each of the L positive and M negative anchor features, respectively. These tailored losses, strategically designed within the framework, enhance the model's ability to distinguish accident instances from normal patterns, paving the way for more robust accident detection in video data.

3) Dilated Temporal Conv-Attention (DTCA) Block: Inspired by attention techniques in video understanding, the proposed DTCA captures multi-resolution local temporal dependencies and global temporal dependencies between video frames. The DTCA block is illustrated in Fig. 4. Multi-scale temporal representations are learned using the CLIP pre-computed features: *F(clip)*. For local temporal dependency, dilated convolutions are employed over the temporal dimension using a pyramid of dilated convolution (*PDC*) [50]. The dilated convolution operation $PDC^{(n)}$, for n equal to 1, 2, and 4, is given by

$$PDC^{(n)} = \text{Conv1D}\left(F(clip), \text{kernel\_size} = 3, \text{dilation} = n\right) \quad (6)$$

Further features from each of *PDC* are processed using RELU non-linearity, batch normalization layer, and dropout layer for regularization.

To capture the global temporal context, the attention block within the transformer [49] is used to process features along the time dimension. Let $F(\text{c})$ denote input features to the attention block, which is convoluted with a 1×1 convolution and a learnable position encoding represent element-wise added to the convoluted features. To reduce spatial dimension, position encoding-added convoluted features are again convoluted using 1×1 convolution. Using these features as $F(c_1)$, $F(c_2)$, and $F(c_3)$, attention block estimates pairwise correlations between features, generating an attention map $M \in \mathbb{R}^{(T\times T)}$ given as

$$M = \text{Softmax}\left(\frac{F(c_1)\cdot F(c_2)^T}{\sqrt{D}}\right)\cdot F(c_3) \quad (7)$$

This attention map M is then matrix multiplied with $F(c_3)$, and convoluted again using 1×1 convolution to get features $F \in \mathbb{R}^{(T\times D)}$. For better global temporal attention, initial features *F(clip)* are divided into three spatial domains: $F(c) \in \mathbb{R}^{(T\times(0\to D/2),(D/2\to D),(0\to D))}$. Each of these features $F(c)$ undergoes separate processing using three separate attention blocks.

Each of the three features processed using attention blocks as well as the three features processed using *PDC* are concatenated and convoluted with a 1×1 convolution. This reduces spatial dimensions from $F \in \mathbb{R}^{(T\times 6D)}$ to $F \in \mathbb{R}^{(T\times D)}$. In addition, these features are processed using RELU non-linearity, batch normalization layer, and dropout layer. A skip connection utilizing the original feature (*F(clip)*) is element-wise added, producing the final temporal feature representation. These features are finally used for selecting the least-K magnitude features of accident and normal video frames. Normal frames, with the least-K magnitude, are selected with a dropout of 0.7 to maintain good generalization. Also, training batches of pairs consisting of an accident and a normal video are processed. After separating the features of normal and accident videos from a pair, a multi-layer perceptron with 3 layers is deployed as a classification head. This classification head outputs a score for each frame of input video out of which K scores, one for each frame feature embedding having the least-K magnitudes are chosen for loss calculation while training.

C. Optimization Procedure

1) Magnitude Enhanced Multiple Instance Learning Loss in Time-VAD: In TIME-VAD, the optimization process employs magnitude-enhanced MIL loss. This loss function, $\mathcal{L}(\mathcal{B}_a, \mathcal{B}_n)$, relies on the discrepancy d_θ. Here, d_θ is calculated as the difference between the mean of the least-K feature magnitude from normal video snippets and abnormal video snippets. Here, the snippet is considered as a frame of the video. The loss is triggered when this difference is less than a margin m. This formulation aims to ensure that the minimum magnitude of snippet feature embedding within the normal video is greater than the minimum magnitude of snippet feature embedding within the abnormal video, enhancing the distinction between these snippet feature embeddings during model optimization. Following RTFM, instead of just taking the least feature magnitude of the snippet's feature, the mean of the least-K feature magnitudes of normal video snippets is compared with the least-K feature magnitudes of abnormal video snippets. This magnitude-enhanced MIL loss is given by

$$\mathcal{L}(\mathcal{B}_a, \mathcal{B}_n) = \max(0, m - d_\theta) \quad (8)$$

where $a_i \in \mathcal{B}_a$ and $n_i \in \mathcal{B}_n$ represent individual frame feature embeddings, and

$$d_\theta = \frac{1}{K}\sum \text{least-K}(\|n_i\|_2) - \frac{1}{K}\sum \text{least-K}(\|a_i\|_2)$$
$$\min(\|n_i\|_2) > \min(\|a_i\|_2)$$

2) Frame Classifier Loss in Time-VAD: Within TIME-VAD, the frame classifier loss is integral to the learning process. This loss, $\mathcal{L}_f(f_\theta, y)$, operates based on the binary cross-entropy formula, summing over a subset of instances designated by the least-K magnitude of frame features, where $y \in (1, 0)$ such that y for least-K magnitude features of snippets from an accident video is 1 and y for least-K magnitude features of snippets from a normal video is 0. Since frame labels are not available directly, we use the least-K magnitude of feature embeddings to match their classification probability with the corresponding video labels. This results in weak supervision on frame level based on video labels. f_θ is a function of x, which is the least-K magnitude of feature embeddings from each of $\mathcal{B}_a$ and $\mathcal{B}_n$. Frame classification loss aims to optimize the model by minimizing the classification error between abnormal and normal snippets based on video labels, enabling effective discrimination between these categories during training. This video label classification loss for the least-K feature magnitude of snippets from accident and normal videos is given as

$$\mathcal{L}_f(f_\theta, y) = \sum_{x\in\text{least-K}} -\big(y\log(f_\theta(x)) + (1-y)\log(1 - f_\theta(x))\big) \quad (9)$$

where x specifically refers to the least-K magnitude feature embeddings selected for classification.

3) Text Anchor-Based Contrastive Loss in Time-VAD: Differentiating itself from traditional supervised contrastive loss [23], TIME-VAD employs the text anchor-based contrastive loss ($\mathcal{L}_{\text{cont}}$). Unlike traditional contrastive learning that includes both attraction and repulsion terms, our formulation focuses solely on attracting relevant features to their corresponding anchors, avoiding explicit repulsion between cross-category pairs to prevent potential interference in the weak supervision setting. Loss function $\mathcal{L}_{\text{cont}}$, leverages the similarity scores between anchor vectors and frame features in x, which are the least-K magnitude of feature embeddings from $\mathcal{B}_a$ and $\mathcal{B}_n$. Here, x is comprised of both accident ($f^{+}_{\text{least-K}}$) and normal ($f^{-}_{\text{least-K}}$) frame feature embeddings, which have the least-K magnitude in the respective accident ($\mathcal{B}_a$) and normal ($\mathcal{B}_n$) video. This process is guided by binary labels (a_{mask} and n_{mask}) for $f^{+}_{\text{least-K}}$ and $f^{-}_{\text{least-K}}$. The text anchor-based contrastive loss ($\mathcal{L}_{\text{cont}}$) is then given by

$$\begin{aligned}\mathcal{L}_{\text{cont}}&(f^{+}_{\text{least-K}}, f^{-}_{\text{least-K}}, a^{+}, a^{-})\\ &= \mathcal{L}_{\text{CL}}(\text{sim}(f^{+}_{\text{least-K}}, a^{+}), a_{\text{mask}})\\ &\quad + \mathcal{L}_{\text{CL}}(\text{sim}(f^{-}_{\text{least-K}}, a^{-}), n_{\text{mask}})\end{aligned} \tag{10}$$

where sim denotes the cosine similarity scores between frame features with least-K magnitude and anchor vectors and binary masks are constructed as follows: a_{mask} are binary labels with the length of 2K that have K entries as 1 for each of least-K magnitude features from an accident video and K entries as 0 for each of least-K magnitude features from a normal video. n_{mask} are binary labels with the length of 2K, which have K entries as 0 for each of least-K magnitude features from an accident video and K entries as 1 for each of least-K magnitude features from a normal video. a^{+} and a^{-} denote the positive and negative anchor vector embeddings, respectively. The video-level label y determines the mask construction, establishing the direct relationship between video labels and text anchor vectors used for contrastive learning.

In addition, the contrastive loss function $\mathcal{L}_{\text{CL}}$ in Eq. (10) encapsulates the probabilistic interpretation of similarity scores between frame features and anchor vectors. The function calculates the mean negative logarithm of the softmax probabilities of similarity scores scaled by a temperature parameter (temp). These probabilities are multiplied by binary masks (mask_i), assigning weights to each score based on whether the instance is labeled as 0 or 1. As discussed, frame masks are made using video labels, which provide weak supervision on the frame level based on video labels. This mechanism enables a nuanced understanding of similarity, facilitating effective separation between accident and normal instances in the contrastive learning process within TIME-VAD. The contrastive loss function $\mathcal{L}_{\text{CL}}$ is given by

$$\begin{aligned}&\mathcal{L}_{\text{CL}}(\text{sim}_i, \text{mask}_i)\\ &= -\text{mean}_{i=1}^{2K}\left(\text{mask}_i \times \log\left(\frac{e^{\text{sim}_i/\text{temp}}}{\sum\left(e^{\text{sim}_i/\text{temp}}\right) \times (1-\text{mask}_i)}\right)\right)\end{aligned} \tag{11}$$

where sim_i represents the similarity score of the i^{th} frame feature, $\text{mask}_i \in \{0, 1\}$ denotes the binary mask, and temp is the temperature parameter used to scale the similarity scores.

4) Total Loss in Time-VAD: For a total loss, temporal smoothness, and sparsity regularisation are also needed as used in [20]. The temporal smoothness ($\mathcal{L}_s(f_\theta, \mathbf{f})$) enforces a similar anomaly score for neighboring snippets. It is given by

$$\mathcal{L}_s(f_\theta, \mathbf{f}) = \sum_{i=1}^{T-1}\left(f_\theta(\mathbf{f}_i) - f_\theta(\mathbf{f}_{i+1})\right)^2 \tag{12}$$

where $\mathbf{f}$ denotes feature embeddings of $(\mathcal{B}_a, \mathcal{B}_n)$. Following the same notations, the sparsity regularisation loss ($\mathcal{L}_r(f_\theta, \mathbf{f})$) imposes a prior in which abnormal events are rare in each abnormal video and is given by

$$\mathcal{L}_r(f_\theta, \mathbf{f}) = \frac{1}{T}\sum_{i=1}^{T} f_\theta(\mathbf{f}_i) \tag{13}$$

The total loss ($\mathcal{L}_t$) is the weighted sum of $\mathcal{L}(\mathcal{B}_a, \mathcal{B}_n)$, $\mathcal{L}_f(f_\theta, y)$, $\mathcal{L}_{\text{cont}}$, $\mathcal{L}_s(f_\theta, \mathbf{f})$, $\mathcal{L}_r(f_\theta, \mathbf{f})$, and is given as

$$\begin{aligned}\mathcal{L}_t &= w_1 \times \mathcal{L}(\mathcal{B}_a, \mathcal{B}_n) + w_2 \times \mathcal{L}_f(f_\theta, y) + w_3 \times \mathcal{L}_{\text{cont}}\\ &\quad + w_4 \times \mathcal{L}_s(f_\theta, \mathbf{f}) + w_5 \times \mathcal{L}_r(f_\theta, \mathbf{f})\end{aligned} \tag{14}$$

where w_j are respective weights for $j \in [1, 5]$. During training, CLIP features were pre-calculated and stored to avoid recalculating them in every epoch, saving GPU memory usage. This is equivalent to freezing the backbone. Text anchors as well as video frame CLIP embeddings are used for training. The total loss is used to train the DTCA block and the classification head, as shown in Fig. 4. During testing, the CLIP backbone, DTCA block, and classification head are frozen, and no loss is calculated. Only video frame CLIP embeddings will be used for testing. The model outputs a score for each frame of the video, representing the frame's accident probability.

TABLE II
ACCIDENT DATASETS USED IN EXPERIMENTS

Dataset	Frames per second	Training samples (positive: P, negative: N)	Testing samples (positive: P, negative: N)	Video length (s)
DAD [51]	20 FPS	455 P, 829 N	165 P, 301 N	5
CCD [45]	10 FPS	1,200 P, 2,400 N	300 P, 600 N	5
DoTA [4]	10 FPS	3,275 P, 4,130 N (N from DAD and CCD)	1,402 P (unaltered test subset)	5 to 30

IV. Implementation and Experimental Results

A. Datasets

Three datasets named car crash dataset (CCD) [45], dash-cam accident dataset (DAD) [51], and detection of traffic anomaly (DoTA) [4] are used in this paper. The key attributes and compositions of these datasets are provided in Table II. CCD and DAD offer unique perspectives on traffic accident videos taken from different cameras, showcasing diverse scenarios and attributes. CCD, with its 10 frames per second (FPS) videos, presents a substantial collection of 1,200 positive and 2,400 negative samples, while DAD's 20 FPS videos comprise 455 positive and 829 negative samples. A comparative analysis between these datasets highlights noteworthy distinctions [46]. CCD predominantly features accidents involving the ego-vehicle (53.4%) with variable appearance features and random accident start time within the last 2 seconds of the video. Contrastingly, DAD videos often depict complex urban traffic accidents where ego-vehicles are not involved (90% of cases) and specifically containing an accident within the last 0.5 seconds. To supplement the scarcity of large-scale annotated datasets, DoTA dataset contains 4,677 videos with varying lengths for accident cases only. However, the proposed method needs both accident and normal videos. Thus in DoTA dataset, 4,130 negative samples for training were formed by combining normal samples from DAD and CCD datasets. The testing subset, comprising of 1,402 positive videos, remains unchanged to maintain consistency with original samples for fair comparison with baselines.

B. Details of Implementation

Instead of generating snippets with multiple frames, we analyzed each frame individually in our experiments. The margin m was consistently set to 100, while K was maintained at 4. In the DTCA architecture, a dropout rate of 0.3 was applied in L1, and its classification head comprised three fully connected perceptron (FCP) layers (sized 512, 128, and 1) separated by dropout layers and ReLU activations. Normal feature selection employed a dropout of 0.7, whereas no dropout was applied for accident feature selection. T was set to 50 in DTCA, which allowed capturing a 5-second temporal context with a 10 FPS dataset. During real-world testing involving cross-dataset evaluation, DAD's 100-frame videos were downsampled to 10 FPS for testing on CCD, resulting in 50 frames, as done in [46]. Conversely, training using the DoTA dataset involved converting variable-length videos to a standard representation of 50 frames through equally spaced frame sampling. During the testing phase, predictions were made using a sliding window approach with 50-frame windows.

Optimizing TIME-VAD involved using the Adam optimizer with a weight decay of 0.0005 and a batch size of 32 instances, each instance combining a normal and an accident video. This paired video sampling in each batch naturally handles video-level class balance during training, while the differential dropout rates (0.7 for normal frames, none for accident frames) along with sparsity regularization loss help address frame-level class imbalance. Learning rates were set at 0.0001 for DAD and DoTA, and 0.0005 for CCD. The loss weights w_1, w_2, w_3, w_4, and w_5 in Eq. (14) were set as 0.0001, 1, 0.002, 0.0008, and 0.008, respectively, to maintain comparable loss magnitudes. In Eq. (11), temp was set as 0.1. Implementation was done using PyTorch, running on Nvidia GeForce GTX 1080 GPU with 12GB memory. During inference, the CLIP model requires 45.23 msec per frame for CLIP feature extraction. Additionally, our model takes 18.82 msec per frame to predict the accident score. Given that human reaction times are not typically this fast, most datasets and analyses are conducted at a rate of 10 FPS. Hence, TIME-VAD can handle this frame rate efficiently. Furthermore, to compare our model with DSTA [46] and GCRNN [45], we used the models provided by the authors on GitHub for calculating metrics that were not directly available in the respective studies. Lastly, we have reported the available metrics of various models from the published results of respective studies for baseline comparison.

C. Evaluation Metrics

The evaluation of a model for accident anticipation involves assessing two critical aspects: correctness and earliness of prediction. While correctness (accuracy) is vital for any VAD system, it is important to note that earliness (timeliness) of detection can not be guaranteed, as it depends on the specific nature of the accident. For example, accidents caused by sudden events like explosions may not be detected at an earlier stage. However, studies like DSTA [46] and GCRNN [45] focused only on the earliness metric, which is a one-sided aspect of evaluation.

1) Correctness Metrics: Under this category, the model's ability to detect accident on frame level is measured using:

Receiver Operating Characteristic - Area Under the Curve (ROC-AUC): In essence, the Area Under the Curve (AUC) quantifies the model's ability to discriminate between positive and negative instances at the frame level [4]. AUC is derived from the ROC curve, which depicts the trade-off between true positive and false positive rates across different classification thresholds. Hence, the AUC serves as a comprehensive measure. Higher AUC values signify more effective model performance, with a perfect model achieving an AUC of 1, while an AUC of 0.5 corresponds to random guessing.

2) Earliness Metrics: These metrics assess the timeliness of accident anticipation and include:

Mean Time-to-Accident (mTTA): This metric assesses the earliness of accident anticipation based on positive predictions [45], [51]. Time-to-Accident (TTA) measures the time taken in seconds, for the system to anticipate an accident at different classification thresholds. For various threshold values, TTA results are obtained. Then, mTTA representing the mean time taken in seconds is calculated as the average of all TTA values across these thresholds. This signifies the average duration between the system's identification of an impending accident and the actual occurrence. Hence, mTTA provides an overall measure of how early the model predicts an accident across different threshold settings.

Average Precision (AP): This metric evaluates the preciseness in predicting future accidents within a video [45], [51]. Using confidence scores, an accident is anticipated if a threshold is reached. True positive predictions occur when the method accurately anticipates an accident before it happens at frame y in the video. Conversely, false positives and false negatives arise when incorrect predictions are made for non-accident and accident videos, respectively. Varying the threshold, precision-recall pairs are formed to plot a precision-recall curve. The average precision is the area under the precision-recall curve that reflects the system's overall accuracy in early anticipation of accidents across the videos. While ROC-AUC provides a general measure of frame-level classification performance, Average Precision (AP) offers a more realistic evaluation for accident anticipation systems. Unlike ROC-AUC which considers all frame predictions, AP only evaluates predictions made before accidents occur, ensuring the metric reflects real-world early warning capabilities. Additionally, AP incorporates time-to-accident weighting, rewarding models that detect accidents earlier rather than just detecting them accurately.

D. Evaluation of the Time-VAD

1) Experimental Analysis of Contrastive Features: Uniform Manifold Approximation and Projection (UMAP) [52] is a dimensionality reduction technique used to visualize

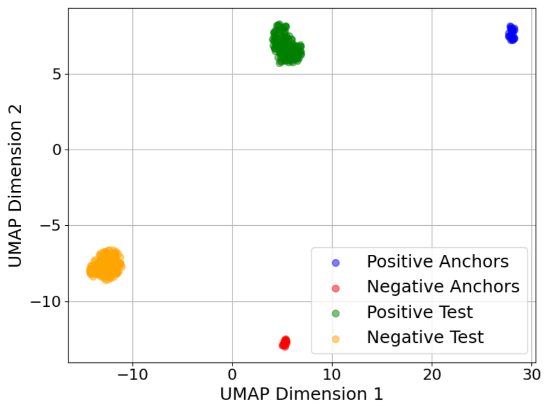

Fig. 5. UMAP plot of CLIP embeddings from 32 randomly selected least magnitude frames of positive (green) and negative (yellow) test videos. Frame embeddings are juxtaposed with positive (blue) and negative (red) text anchors.

TABLE III

COMPARISON OF ROC-AUC SCORES BASED ON ACCIDENT AND NORMAL FRAME EMBEDDING MAGNITUDES

Dataset	ROC-AUC **(Accident frame magnitude is considered maximum)** (%)	ROC-AUC **(Accident frame magnitude is considered minimum)** (%)
DAD [51]	75.36	**84.43**
CCD [45]	**95.5**	95.38
DoTA [4]	64.26	**94.44**

TABLE IV

COMPARATIVE ANALYSIS OF ACCIDENT ANTICIPATION MODELS ON CCD DATASET

Method	ROC-AUC(%)
DSTA [46]	53.19
GCRNN [45]	47.87
TIME-VAD (Maxima for ROC-AUC)	**95.38**
TIME-VAD (Maxima for AP)	91.00

high-dimensional data in a lower-dimensional space while maintaining its inherent structure and relationships. This study applied UMAP to visualize embeddings generated by the TIME-VAD model trained on CCD. The UMAP plot, as depicted in Fig. 5, showcases embeddings obtained from randomly selected 32 positive and 32 negative test videos. Considering the least four magnitudes of frame embeddings from each video, the total embeddings constitute 128 positive test embeddings and 128 negative test embeddings. These frame embeddings were then contrasted with 27 positive and 19 negative text anchors. The UMAP visualization demonstrates clear clustering patterns that support our approach. As shown in Fig. 5, while text anchors and their corresponding frame embeddings may appear separated in this two-dimensional projection, they follow consistent relational trends across both UMAP dimensions. Specifically, positive text anchors and positive test frame embeddings form distinct clusters that are well-separated from the clusters of negative text anchors and negative test frame embeddings. The positive frame samples consistently maintain higher values in both UMAP dimensions compared to negative frame samples, following the same pattern observed between positive and negative text anchors. This parallel alignment of trends, rather than absolute proximity in the reduced space, validates our contrastive learning approach's effectiveness in meaningfully organizing the embedding space and establishing proper relationships between the magnitude of textual and visual embeddings for positive and negative cases.

2) Experimental Analysis for Magnitude Enhanced Feature Learning: An experimental analysis was conducted to explore the impact of considering the magnitude of accident and normal frame instances on the ROC-AUC metric across three datasets: DAD, CCD, and DoTA. For each of the datasets, the model is trained and then tested on the test set of the respective dataset. Experiments were performed for two different cases on each dataset. In the first case, it was assumed that accident frames tend to have larger magnitude embeddings, while normal traffic frames exhibit smaller magnitude embeddings. In the second case, it was assumed that accident frames tend to have smaller magnitude embeddings, while normal traffic frames exhibit larger magnitude embeddings. Results of the ROC-AUC metric for the different cases are enlisted in Table III. The results depict that the second assumption led to significant improvements in ROC-AUC metrics for the relatively challenging DAD and DoTA datasets: from 75.36% to 84.43% for DAD and from 64.26% to 94.44% for DoTA dataset. Conversely, the relatively simpler dataset, CCD, showed a marginal change in performance: ROC-AUC remained comparably high regardless of the approach used, scoring 95.5% and 95.38% in the respective cases.

3) Comparison to the State-of-the-Art Models: We performed an experimental analysis on the CCD dataset that demonstrates the effectiveness of our method when compared with other accident anticipation models based on frame-level ROC-AUC metric. As presented in Table IV, among the assessed methods, the proposed TIME-VAD model significantly outperforms existing models. TIME-VAD model showcases a notable ROC-AUC score of 95.38%, while other methods, such as DSTA and GCRNN, show relatively lower performance with ROC-AUC scores of 53.19% and 47.87%, respectively. A notable observation from the training behavior is the trade-off between early anticipation (measured by AP) and correct anticipation (measured by frame-level ROC-AUC). Models situated on the Pareto-optimal frontier strike an efficient balance, indicating that enhancing the AP often comes at the expense of a decrease in frame-level ROC-AUC [53]. However, even when we optimized for maximizing AP, the ROC-AUC metric did not fall too much and was significantly higher than the existing methods. Selecting an optimal model for implementation should align with specific user requirements, which account for diverse environmental factors influencing accident anticipation systems.

Our experimental analysis on the DoTA dataset emphasizes the superior performance of the proposed TIME-VAD model in terms of frame-level ROC-AUC score, as enlisted in Table V. In comparison to a spectrum of unsupervised, weak-supervised, and supervised methods, our model consistently outperforms the previous methods. Specifically, when considering unsupervised methods such as ConvAE [5] and ConvLSTMAE [54] with both gray and flow inputs, as

TABLE V
COMPARATIVE ANALYSIS OF ACCIDENT ANTICIPATION MODELS ON DoTA DATASET. TEXT INPUT IS ONLY USED FOR TRAINING FOR ALL METHODS

Method	Training type	Input	ROC-AUC(%)
ConvAE (gray) [5]	Unsupervised	Gray	64.3
ConvAE (flow) [5]		Flow	66.3
ConvLSTMAE [54] (gray)		Gray	53.8
ConvLSTMAE [54] (flow)		Flow	62.5
AnoPred [7]		RGB	67.5
AnoPred [7] + Mask		Masked RGB	64.8
TAD [18]		Box + Flow	69.2
TAD [18] + ML [10], [55]		Box + Flow	69.7
Ensemble [4]		RGB + Box	73.0
STFE [66]		RGB + Box	79.3
RTFM (DTCA-based) [39]	Weak-supervised	RGB + Text	64.26
RTFM [39] with M1 [67]		RGB	57.9
RTFM [39] with M2 [67]		RGB	56.0
RTFM [39] with M3 [67]		RGB	78.2
MGFN [68] with M1 [67]		RGB	66.6
MGFN [68] with M2 [67]		RGB	67.4
MGFN [68] with M3 [67]		RGB	67.4
URDMU [69] with M1 [67]		RGB	57.5
URDMU [69] with M2 [67]		RGB	54.8
URDMU [69] with M3 [67]		RGB	73.0
OE-CTST [70] with M1 [67]		RGB	70.9
OE-CTST [70] with M2 [67]		RGB	71.9
OE-CTST [70] with M3 [67]		RGB	75.6
TIME-VAD (ours)		RGB + Text	**94.44**
FC	Supervised	RGB	61.7
LSTM [56]		RGB	63.7
Encoder-Decoder [57]		RGB	73.6
TRN [44]		RGB	78.0
VST + TCN [72]		RGB	72.7
CTFMN [72]		RGB	78.5
MOVAD [71]		RGB	82.17
TTHF [65]		RGB + Text	84.7

well as the AnoPred model [7], our proposed TIME-VAD showcases significant improvements. Moreover, even against supervised methods like fully connected network (FC), LSTM [56], Encoder-Decoder [57], and TRN [44], the TIME-VAD model exhibits remarkable advancement, securing a substantial ROC-AUC score of 94.44%. Notably, the TIME-VAD model surpasses the performance of unsupervised and supervised approaches, proving its efficacy in accident anticipation, and demonstrating its potential as a reliable solution in this domain.

Furthermore, using the CCD dataset, our experiments focused on gauging the model's readiness or earliness to anticipate accidents by evaluating two critical metrics: AP and mTTA, as displayed in Table VI. Notably, among the various models and their respective performance scenarios, case (b) of TIME-VAD showcased the highest values for both AP and mTTA. Compared to a recent method (W3AL) [73] that achieves AP of 99.7% with 3.93 mTTA, TIME-VAD (b) outperforms it by reaching 99.97% with 4.99 mTTA, though such saturation of high AP necessitates frame-level ROC-AUC evaluation for meaningful comparison. This underlines the proposed model's superior readiness to anticipate accidents at an earlier stage. Also, when compared with case (b) of TIME-VAD model, earliness metrics for DSTA and GCRNN

TABLE VI
COMPARATIVE ANALYSIS FOR EARLINESS METRICS WITH CORRESPONDING CORRECTNESS METRIC ON CCD DATASET

Method	AP(%)	mTTA(s)	ROC-AUC(%)
DSA [51]	99.6	4.52	NA
DSTA [46]	99.6	4.87	53.19
GCRNN [45]	99.54	4.74	47.87
GSC [74]	99.3	3.58	NA
CRASH [75]	99.6	4.91	NA
W3AL [73]	99.7	3.93	NA
LATTE [76]	98.77	4.53	NA
TIME-VAD (a) (Maxima for ROC-AUC)	77.17	4.07	**95.38**
TIME-VAD (b) (Maxima for AP)	**99.97**	**4.99**	91.00
TIME-VAD (c) (Best for both applications)	99.89	3.55	92.03

models are less but comparable. However, ROC-AUC score is significantly low for both DSTA and GCRNN models in comparison to the TIME-VAD. For video level analysis at 80% recall, TIME-VAD's anticipation-focused variant (a) achieves TTA of 2.92s with precision of 0.78. The detection-focused variant (b) demonstrates higher precision (1.0) with longer TTA of 4.86s at 80% recall, illustrating the effective trade-off between early warning and detection accuracy.

As discussed, higher ROC-AUC values might result in lower earliness-related metrics, reflecting a trade-off between the earliness metric and the correctness metric (frame-level ROC-AUC metric). Interestingly, observations from DSTA [46] and GCRNN [45] models suggest potential overfitting issues. These issues are specifically due to learning from accident-irrelevant cues within the scene. This can be the main reason for the significantly low ROC-AUC for both DSTA and GCRNN models. In response, we developed a more balanced model denoted as case (c) in Table VI, aiming to achieve a harmonious performance across all three metrics. To achieve the TIME-VAD model case (c), we increased the weight of frame classifier loss ($\mathcal{L}_f(f_\theta, y)$) to 10 and then finetuned the model of the case (a) in Table VI. This balanced approach in the TIME-VAD model, where all three metrics, AP, mTTA, and ROC-AUC demonstrate commendable performance, showcases its potential as a promising solution for accurate and timely accident anticipation. To further validate the findings, a detailed analysis of precision-recall characteristics is conducted across different threshold values for all three TIME-VAD variants, as shown in Fig. 6. The comprehensive evaluation reveals distinct performance characteristics, with TIME-VAD (a) demonstrating superior overall performance by achieving the highest area under precision-recall curve (PR-AUC) (0.703) and ROC-AUC (0.954), reaching peak precision values of around 0.95 at higher thresholds while maintaining robust recall, validating its optimization for detection accuracy. TIME-VAD (b), optimized for early anticipation, exhibits lower but still competitive performance (PR-AUC: 0.379, ROC-AUC: 0.910), reflecting the inherent trade-offs when prioritizing temporal anticipation over pure detection accuracy. TIME-VAD (c) achieves intermediate performance (PR-AUC: 0.538, ROC-AUC: 0.920), successfully

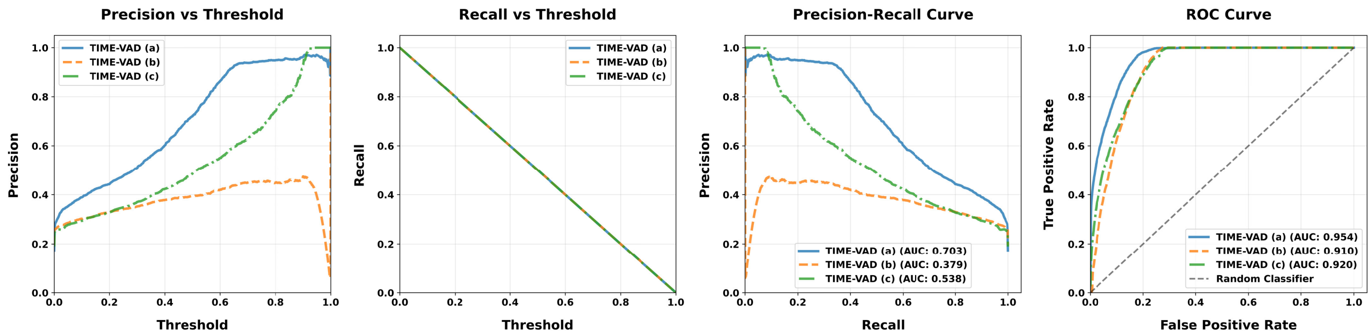

Fig. 6. Comprehensive performance analysis of TIME-VAD variants on the CCD dataset showing (from left to right): Precision vs threshold, Recall vs threshold, Precision-Recall curve, and ROC curve. Three optimization strategies are compared: TIME-VAD (a) optimized for detection accuracy, TIME-VAD (b) optimized for early anticipation, and TIME-VAD (c) as a balanced approach fine-tuned from variant (a).

TABLE VII

ABLATION STUDY RESULTS FOR DIFFERENT COMBINATIONS OF LOSS COMPONENTS FOR CCD DATASET. ✓ INDICATES INCLUSION WHILE × INDICATES OMISSION OF RESPECTIVE LOSS COMPONENT

$\mathcal{L}_f(f_\theta, y)$	$\mathcal{L}(\mathcal{B}_a, \mathcal{B}_n)$	$\mathcal{L}_{\text{cont}}$	ROC-AUC(%)	mTTA (s)	AP(%)
✓	×	×	91.65	4.83	98.43
×	✓	×	80.47	4.30	**99.81**
×	×	✓	18.19	3.52	18.96
✓	✓	×	93.56	4.60	96.20
✓	×	✓	71.78	4.33	46.34
×	✓	✓	93.63	**4.95**	94.09
✓	✓	✓	**95.38**	4.07	77.17

balancing between the two optimization strategies. Notably, all variants maintain strong discriminative ability with ROC-AUC values above 0.91, while the precision-recall analysis reveals more nuanced differences in handling class imbalance, with the threshold sensitivity analysis demonstrating that TIME-VAD (a) maintains the most stable precision-recall trade-offs across different operating points, validating the effectiveness of the framework.

4) Comparative Analysis of Temporal Models: Different temporal modelling methods to capture frame events are compared by replacing DTCA in TIME-VAD architecture on CCD test dataset. Also, the inference time per frame from the temporal method part only is noted. The DTCA model achieved the highest ROC-AUC of 95.38% with an inference time of 2.84ms per frame. The Long Short-Term Memory (LSTM) model with two bidirectional layers and 384 hidden units per direction reached 94.12% ROC-AUC but required 3.75ms per frame. Temporal Convolutional Network (TCN) with three stacked dilated convolutional layers (dilation rates 1, 2, 4) demonstrated good efficiency with 91.73% ROC-AUC and the fastest inference time of only 0.85ms. The Transformer architecture consisting of three multi-head attention blocks (8 heads each) matched TCN's detection capability with 91.72% ROC-AUC and required 1.37ms per frame. Convolutional LSTM (Conv-LSTM), combining two convolutional layers (dilations 1, 2) with a two-layer bidirectional LSTM, achieved 89.58% ROC-AUC but had the highest computational cost at 5.59ms per frame.

5) Ablation Study for Loss: To evaluate the effectiveness of different components of our loss function, we conducted an ablation study by keeping the temporal smoothness and sparsity regularization losses constant and varying the presence of other losses, namely, text anchor-based contrastive loss ($\mathcal{L}_{\text{cont}}$), frame classification loss ($\mathcal{L}_f(f_\theta, y)$), and magnitude enhanced multiple instance learning loss ($\mathcal{L}(\mathcal{B}_a, \mathcal{B}_n)$). Our goal is to optimize the ROC-AUC metric for CCD dataset. We reported three different metrics in Table VII: ROC-AUC, mTTA, and AP. The values for the ROC-AUC metric, summarized in Table VII, reveal that the combination of all three losses resulted in the highest ROC-AUC of 95.38%. The use of only text anchor-based contrastive loss led to significantly lower performance across all metrics. Combinations including the magnitude enhanced multiple instance learning loss generally performed well, particularly when combined with other losses. The inclusion of frame classification loss and magnitude enhanced multiple instance learning loss together yielded better results compared to when either was used alone. This ablation study demonstrates the importance of combining different loss components to enhance accuracy and robustness.

6) Experiment for Assessing Model's Generalizability: Experiments were conducted to assess the model's ability to generalize by training on the CCD dataset and testing on the DAD test subset, simulating real-world applications. The results are enlisted in Table VIII. In these cross-domain evaluations, the readiness of models for accident anticipation in uncontrolled settings was scrutinized. During this rigorous testing, DSTA's performance significantly declined for accident earliness-related metrics (AP). Similarly, our model exhibited limited efficacy in predicting accidents early. However, comparative analysis revealed that our model outperformed DSTA in the generalization experiment. This is demonstrated as AP in DSTA decreased by 64.12% while for TIME-VAD the decrement is 55.38%.

Remarkably, our model showcased improved performance in terms of ROC-AUC, demonstrating its enhanced adaptability and robustness when tested in real-world scenarios. In Table VIII, the underlined metrics highlight the primary assessment parameters for a given model, emphasizing the notable variations in model performance between controlled training settings and real-world applications. The findings

TABLE VIII

Testing in Real-World Scenario. Training Is Done on CCD and Testing Is Done on DAD Test-Set. Relevant Metrics for Each Method Are Underlined. AP and ROC-AUC Are in % and mTTA Is in s

Method	Result for testing on CCD for CCD trained method	Result for testing on DAD for CCD trained method	Remark
DSTA [46]	ROC-AUC: 53.19 mTTA: 4.87 AP: 99.6	ROC-AUC: NA mTTA: 4.99 AP: 35.74	AP decreased by 64.12%
TIME-VAD (Maxima for AP)	ROC-AUC: 91.00 mTTA: 4.99 AP: 99.97	ROC-AUC: 46.81 mTTA: 4.392 AP: 44.61	AP decreased by **55.38%**
TIME-VAD (Maxima for ROC-AUC)	ROC-AUC: 95.38 mTTA: 4.07 AP: 77.17	ROC-AUC: 95.50 mTTA: 4.226 AP: 21.53	ROC-AUC increased by **0.12%**

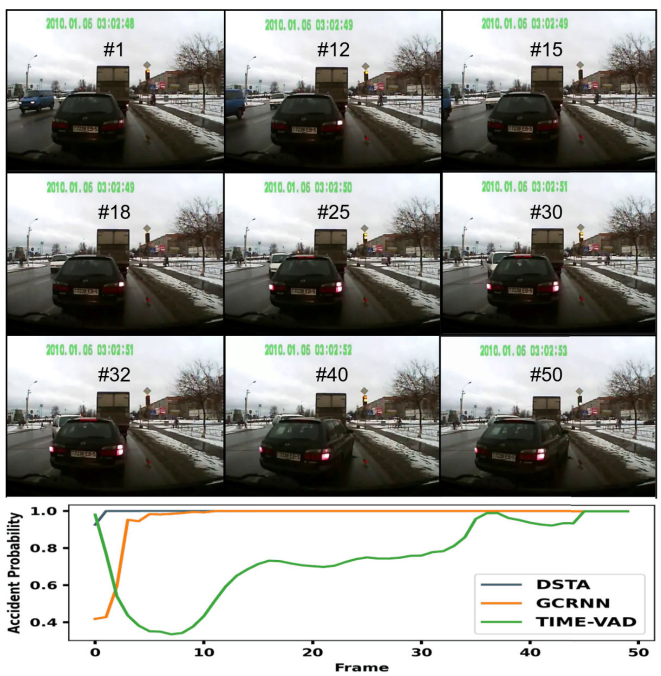

Fig. 7. Qualitative comparison of accident probability for an accident video with accident event starts from the 30th frame onwards. Selected frames with numbers are presented.

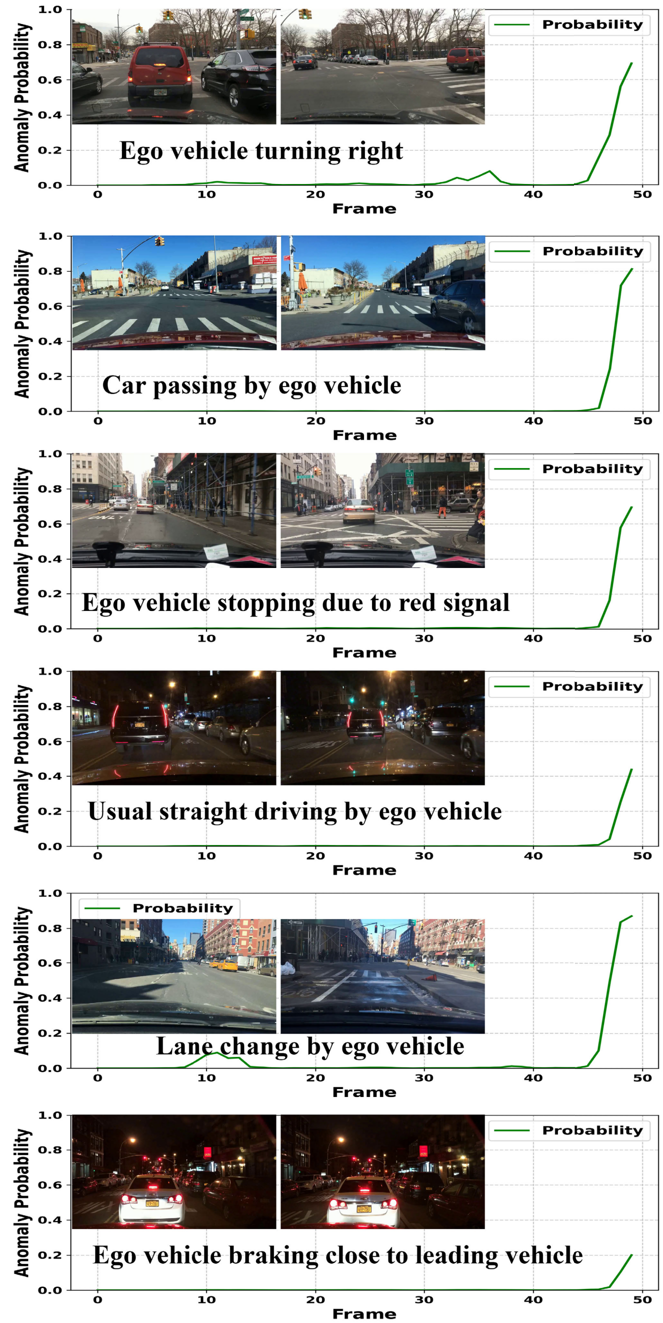

Fig. 8. Qualitative analysis of accident probability for normal videos by TIME-VAD. First and last frames are shown for each plot.

suggest that while our model presents promising capabilities in generalization, there is a need for further optimization to bolster its performance in accident earliness metrics when deployed in uncontrolled environments. However, as pointed out, earliness metrics are dependent on the nature of the dataset that captures relevant vehicles. This is not the case in the DAD dataset, as the aforementioned ego-vehicles are not present in most of the samples along with hard samples of accident cases. Thus, for DAD the earliness metrics testing has moderate performance with AP of 44.61%.

7) Qualitative Results: Qualitative analysis of different models unveils critical insights into their behavior and the cues they prioritize in accident anticipation. Notably, when presented with a challenging sample from the CCD dataset, as shown in Fig. 7, DSTA [46] and GCRNN [45] exhibited signs of over-fitting, displaying a sharp inclination in accident probability right from the video's onset. This inclination, however, seemed more influenced by scene characteristics than actual accident cues, indicating a susceptibility to learning from accident-irrelevant features. In contrast, our model showcased a more nuanced and consistent prediction approach. On a close examination of the video, the imminent accident seems perceptible around the 15th frame, and the actual incident occurs on the 30th frame. Our model consistently gauged the evolving likelihood of an accident occurrence, demonstrating a more accurate and tempered assessment compared to the overtly biased predictions observed in the other models.

Furthermore, as illustrated in Fig. 8, the TIME-VAD maintains consistently low accident probabilities across various normal driving scenarios, including routine maneuvers such as turning, lane changes, and stopping at red signal. While some scenarios show elevated accident probabilities toward the end of the sequences due to potentially risky maneuvers, all videos are correctly classified as normal when using a high threshold of 0.9, demonstrating the model's robust discrimination capability while maintaining high precision. This robust performance on non-accident scenarios further validates our model's ability to discriminate between genuine accident precursors and normal driving events.

V. Conclusion

In this study, we introduced TIME-VAD, a model based on text-informed feature magnitude enhancement using contrastive multiple-instance learning. Through extensive evaluation across diverse benchmark datasets, TIME-VAD outperforms existing state-of-the-art models in both accuracy and early accident detection. This innovative framework exhibits superior discriminative capabilities, particularly in identifying subtle anomalies and ensuring sample efficiency crucial for traffic video analysis. The inclusion of the DTCA block further refines temporal attention, enhancing the model's ability to perceive nuanced accident risks within video frames. Our study highlights the effectiveness of TIME-VAD, showcasing an accuracy of 94.44% (measured by ROC-AUC) on the challenging DoTA dataset, which has the largest sample size. Additionally, for the smaller sample size dataset of DAD, TIME-VAD achieves an accuracy of 84.43%. In terms of generalization, when the model is trained with CCD, a dataset of moderate sample size, and tested on DAD, the accuracy for DAD reaches 95.50%, surpassing the CCD test accuracy by 0.12%. Furthermore, TIME-VAD excels in early accident detection, achieving AP of 99.97% and an mTTA of 4.99 seconds on the CCD dataset. This work contributes to establishing a valuable foundation for advancing accident anticipation systems. It also showcases the potential of multi-modal integration for proactive safety enhancement in transportation systems.

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Sumit Mishra received the B.Tech. degree in electronics and communication engineering from Dr. A.P.J. Abdul Kalam Technical University, India, in 2016, and the M.S. degree from the Robotics Department, Korea Advanced Institute of Science and Technology, South Korea, in 2023, where he is currently pursuing the Ph.D. degree in robotics. Earlier, he was with Learnogether Technologies Pvt. Ltd., as a Research Consultant. His research interests include vision-language models, multimodal foundation models, cross-modal alignment by deep learning, explainable AI for safety-critical systems, and computer vision for intelligent transportation systems.

Medhavi Mishra received the B.Tech. degree in production and industrial engineering from the Motilal Nehru National Institute of Technology, Allahabad, India, in 2016, and the M.S. degree in mobility from Korea Advanced Institute of Science and Technology, South Korea, in 2025, where she is currently pursuing the Ph.D. degree in mobility. Earlier, she was with HighRadius Technologies Pvt. Ltd., as a Techno-Functional Consultant. Her research interests include traffic simulation and modeling, multi-modal perception, and deep learning applications in intelligent transportation systems.

Pranjay Shyam received the B.Tech. degree in electronics and communication engineering from Shiv Nadar University (SNU), India, in 2016, the M.Sc. degree in electrical engineering from the Technical University of Denmark (DTU), and the M.S. and Ph.D. degrees from the Mechanical Engineering Department, Korea Advanced Institute of Science and Technology (KAIST), South Korea, in 2018 and 2022, respectively. He is currently an Applied Scientist with Amazon Fulfilment Technologies and Robotics, USA. His research interests include multi-modal scene reconstruction and understanding.

Dongsoo Har (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in electronics engineering from Seoul National University, and the Ph.D. degree in electrical engineering from Polytechnic University, Brooklyn, NY, USA. He is currently a Faculty Member with KAIST. He has authored and published more than 100 papers in international journals and conferences. He has also presented invited talks and keynote in international conferences. His research interests include optimization of communication system operation and transportation system development with embedded artificial intelligence. He was a member of the Advisory Board and the program chair, the vice chair, and the general chair of international conferences. He was a recipient of the Best Paper Award from *ETRI Journal* and the Best Paper Award (Jack Neubauer Award) from IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY.